

ORBIT

Team playbook: foundation reference

AI-powered global intelligence on a 3D globe. Global → Country → Event → Explanation → Ask ORBIT.

Team: Arham · Ayman · Affan · Shrey · Hardik

Generated 2026-09-19 from the Markdown sources in the repository. If this PDF and the repository disagree, the repository wins.

ORBIT team playbook

The foundation knowledge for the five ORBIT developers, in one place. Detailed rules live in the documents linked from each section. This playbook summarises them and never overrides them.

Document	What it is the authority on
AGENTS.md	Rules for people and coding agents: stack, folder ownership, contracts, styling, testing, Git
tasks.json	Who does what: every task, owner, branch, dependencies, acceptance criteria
docs/API.md	API routes, request/response shapes, data dictionary, mock data format
docs/INTEGRATION.md	Branches, commits, PRs, contract changes, merging
docs/DESIGN_SYSTEM.md	Tokens, primitives, UI states and conventions
docs/ALLOCATOR.md	The personal LLM prompt that tells each person what to do

1. Project overview

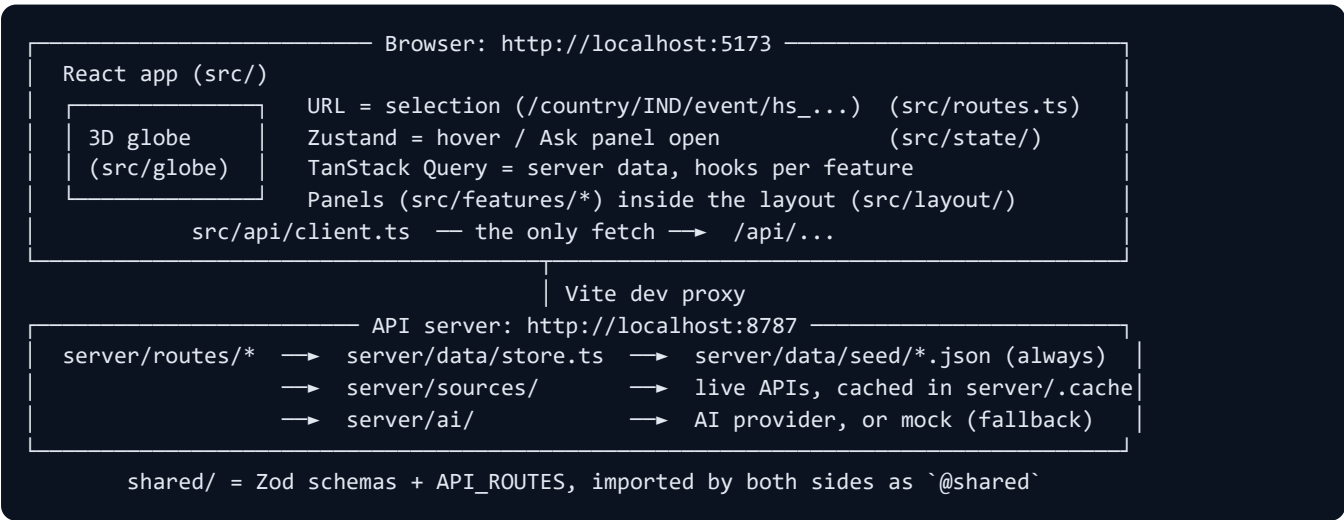
ORBIT is an AI-powered global intelligence platform. It combines geopolitical intelligence, healthcare intelligence, global news, geography, history/timelines and explainable AI around an interactive 3D globe.

Flow: Global → Country → Event → Explanation → Ask ORBIT.

Look: deep black, electric cyan, amber/orange. Futuristic, premium, cinematic and clean.

Constraints: 24-hour hackathon, five student developers, live demo on one laptop, must survive bad wifi. Our priorities, in order: reliability, speed, visual quality, easy collaboration, few merge conflicts, maintainability, demo quality.

2. Architecture



The reasons behind each part:

- **One small server:** it keeps API keys secret, avoids browser CORS blocks, and caches responses for the offline demo.
- **Seed data first:** the demo never depends on wifi. Live data only adds to it.
- **The mock AI** works with no key and is the automatic fallback.
- **No database:** the data is mostly read-only and small.

3. Stack (frozen)

Vite 8 · React 19 · TypeScript (strict) · react-globe.gl / three.js · Express 5 (tsx) · Zod 4 · TanStack Query 5 · Zustand 5 · React Router 7 · CSS Modules + tokens · Vercel AI SDK (ai , @ai-sdk/google) · Vitest 3 + Testing Library · ESLint 9 + Prettier · **npm** · **Node ≥ 22.12** (team uses 24). Details and forbidden alternatives: AGENTS.md section 2.

4. Repository map

AGENTS.md	CLAUDE.md	.cursor/rules/	Agent rules (AGENTS.md is the source)
README.md			Start here
tasks.json			Task manifest (who does what)
.env.example			Every environment variable, safe defaults
docs/			API, integration, design system, playbook, allocator, PDF
shared/			Contracts: schemas/ (one file per entity) + api.ts (routes,
envelope)			
server/			
index.ts	app.ts	env.ts	http.ts
routes/			
(insights, ask)			
data/store.ts	data/seed/*.json		
sources/			
ai/			
src/			
App.tsx	routes.ts	main.tsx	
api/client.ts			
layout/			
globe/			
features/<name>/			
ui/			
styles/			
state/			
test/			
public/data/countries.geojson			
scripts/			
mcp/	workshop-data/	.vscode/mcp.json	

5. Contracts

The rules, summarised (full rules in AGENTS.md section 4 and docs/API.md):

- Every data type exists **once** as a Zod schema in shared/schemas/. The TypeScript type comes from z.infer.
- Import from @shared only. Never redefine a type. OrbitEvent, never Event.
- IDs: country = IND. Others = evt_, hs_, news_, src_, tl_, ins_ + lowercase slug.

- Dates are ISO UTC strings. Coordinates are {lat, lng}. Severity is an integer 1–5. Fields are camelCase. Enums are lowercase.
- Responses: { data } or { error: { code, message } }.
- Changing a contract → docs/INTEGRATION.md#contract-changes.

6. API contract and data dictionary

Route	Returns
GET /api/health	server status, data mode, AI provider
GET /api/countries, /api/countries/:id	Country list / one Country
GET /api/countries/:id/events, /api/countries/:id/timeline	a country's events / history
GET /api/events?kind=, /api/events/:id	all events (optionally by kind) / one event
GET /api/news?countryId=	headlines
GET /api/sources	sources
GET /api/insights/:subjectType/:subjectId	AI explanation for global/world, country/IND or event/<id>
POST /api/ask	AI answer to { question, context? }

Entity	Key fields
Country	id, name, region, capital, centroid, population, summary, riskLevel
OrbitEvent	kind (geopolitical health), id, title, summary, countryIds, location, occurredAt, severity, sourceIds, tags, plus kind-specific fields
NewsHeadline	id, title, url, sourceId, publishedAt, countryIds, eventId?
Source	id, name, url, type, reliability
TimelineEvent	id, countryId, date, datePrecision, title, description, eventId?
AllInsight	id, subjectType, subjectId, summary, keyPoints, sourceIds, confidence, provider, generatedAt

Full shapes, errors, relationships and the mock-data format: docs/API.md.

7. Environment setup

```
# once
git clone https://github.com/Ayman-Alam-India/orbit C:\dev\orbit # NOT inside OneDrive
cd C:\dev\orbit
node -v # must be 22.12 or newer
npm install
copy .env.example .env # optional: every value has a safe default

# every day
npm run dev # web on http://localhost:5173 + API on :8787
npm run check # lint + tests + build: must pass before any PR
```

Variable	Default	Meaning
PORT	8787	API server port (Vite proxies <code>/api</code> to it)
DATA_MODE	mock	mock = seed only, live = seed + live sources
AI_PROVIDER	mock	mock or google
GOOGLE_GENERATIVE_AI_API_KEY	empty	Only in your local <code>.env</code> , never committed

8. Design system

Tokens in `src/styles/tokens.css` : colour, severity scale, typography, spacing, radii, glows, motion, layers.
Primitives in `src/ui` : Panel, QueryState, Loader, ErrorState, ErrorBoundary, ItemList, SeverityBadge. Rule: CSS Modules + tokens only, no raw colours. Details: `docs/DESIGN_SYSTEM.md`.

9. Team ownership

Only **Arham's** strength (frontend) was established during planning. The other roles follow the workload split the team chose (Hardik and Affan heavier, Ayman medium, Shrey lighter), not assumed skills. If a role doesn't fit someone, swap owners in `tasks.json` and `AGENTS.md` together (Hardik).

Person	Primary responsibility	Owens	Depends on	Hands off to
Hardik (heavy)	Integrator: merges, contracts, API server, live data, demo hardening	<code>shared/</code> , <code>server/</code> core + <code>routes/</code> + <code>sources/</code> + <code>store.ts</code> , <code>src/App.tsx</code> , <code>src/routes.ts</code> , <code>src/api/</code> , configs, <code>package*.json</code> , docs, <code>tasks.json</code>	Everyone's PRs	Everyone (contract updates, merged main)
Affan (heavy)	AI: real provider, explanations, Ask ORBIT	<code>server/ai/</code> , <code>server/routes/ai.ts</code> , <code>src/features/insight/</code> , <code>src/features/ask/</code>	Hardik (contract changes), Arham (primitives, layout)	Ayman/Arham (InsightCard), Hardik (AI cache for demo)
Arham (medium-heavy)	Design system, app shell, cinematic globe, global view	<code>src/styles/</code> , <code>src/ui/</code> , <code>src/globe/</code> , <code>src/layout/</code> , <code>src/state/</code> , <code>public/</code>	Shrey (data density), Hardik	Everyone (primitives)
Ayman (medium)	Country, event and timeline views	<code>src/features/country/</code> , <code>src/features/event/</code> , <code>src/features/timeline/</code>	Arham (primitives), Affan (InsightCard), Shrey (data)	Arham (layout fit)
Shrey (light)	Curated data, news and health panels, demo script	<code>server/data/seed/</code> , <code>src/features/news/</code> , <code>src/features/health/</code> , <code>docs/DEMO.md</code>	Hardik (contracts)	Everyone (data), Hardik (demo script)

Shared files that need coordination: see `AGENTS.md` section 3 ("Shared hot files").

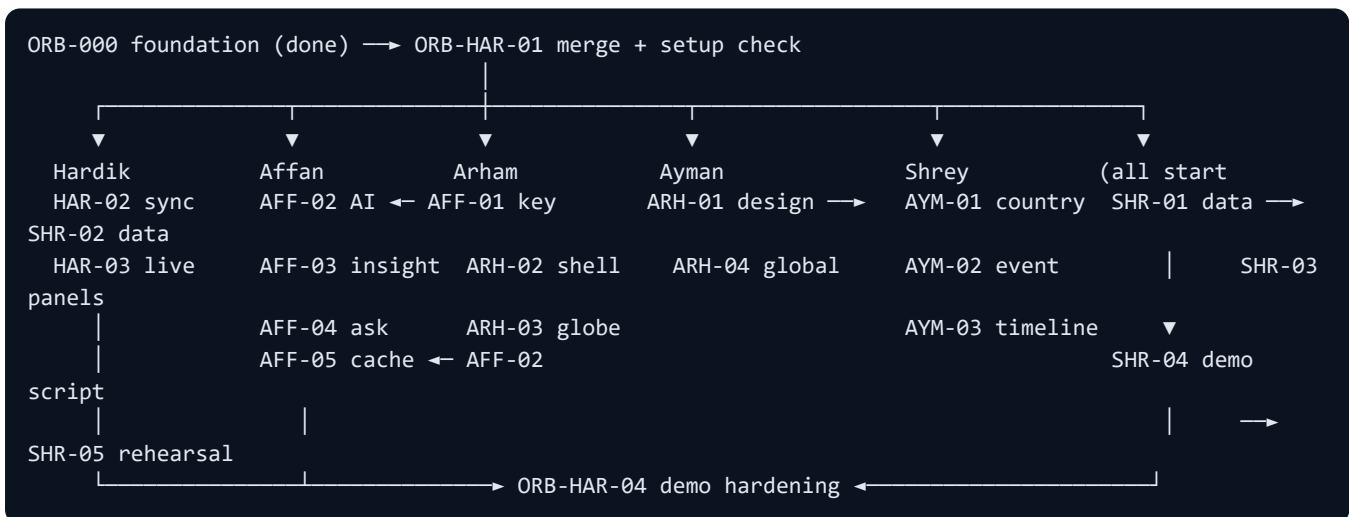
10. Task system

`tasks.json` is the only list of work. Each task has:

Field	Meaning
id	ORB-<PERSON>-<NN> (e.g. ORB-ARH-03). ORB-000 = foundation
owner	The one person responsible
status	todo , in_progress or done . Hardik updates it on merge (tasks.json is a shared hot file)
priority	P0 demo breaks without it · P1 should have · P2 nice to have
targetWindow	Hours from the start (H0) when it should happen
branch	The Git branch to use
paths	The only files/folders the task may change
dependsOn	Tasks that must be merged first
acceptance	What "done" means
tests	Tests to add or keep green
handoff	Who to tell what when you finish

Work order for each person: P0 before P1 before P2, and within a priority, the earliest `targetWindow` whose `dependsOn` tasks are done. `npm test` checks that `tasks.json` has valid IDs, owners and dependencies, with no cycles.

11. Dependency graph



Right after ORB-HAR-01, **all five people can work in parallel**. ORB-AFF-01 (getting the AI key) can start at H0.

12. Git workflow (summary)

Branch from `main` using the task's `branch` name, then make small commits (`area: change`). Before the PR, `git pull origin main` and run `npm run check` . The PR title is `[ORB-XXX-NN] summary` . Hardik squash-merges. Sync points happen every ~4h. Full rules: `docs/INTEGRATION.md` .

13. Integration workflow

1. Contract/API changes merge first (Hardik) → everyone pulls.
2. Server changes merge next, then UI.
3. After each merge batch, Hardik runs `npm run check` + the click-through on `main`.
4. A broken `main` is fixed or reverted before new work continues.

14. Testing

Layer	Tests (existing)	What they protect
Contracts	<code>shared/contracts.test.ts</code>	ID, date, severity and event-kind rules
Data	<code>server/data/seed.test.ts</code>	Every seed file is valid and every reference exists
API	<code>server/app.test.ts</code>	Every route returns contract-valid <code>{ data }</code> / <code>{ error }</code>
Frontend	<code>src/api/client.test.ts</code>	Envelope unwrapping, error codes
Integration	<code>src/App.test.tsx</code>	The real UI rendering real API data (globe mocked)
Manifest	<code>scripts/tasks.test.ts</code>	<code>tasks.json</code> IDs, owners and dependencies

Run all tests with `npm test` (Or `npm run test:watch`). The gate is `npm run check`. Each task adds the tests listed in `tasks.json`.

15. Deployment plan

There is no cloud deployment. The demo runs with `npm run dev` on one laptop (ORB-HAR-04):

1. Pull the final `main`, `npm install`, `npm run check`.
2. Create the demo `.env` (`AI_PROVIDER=google` + key, `DATA_MODE=live` to warm caches).
3. Click through the full demo once online (fills `server/.cache/`), then **turn wifi off** and run it again.
4. Keep the backup recording (ORB-SHR-05) ready.

`npm run build` + `npm run preview` produces a static frontend build, but the API server is still needed, so we demo with `npm run dev`.

16. Coding-agent rules

`AGENTS.md` is read automatically by Codex and Cursor. `CLAUDE.md` imports it for Claude Code, and `.cursor/rules/orbit.mdc` points Cursor to it. The key rules:

- Work on one task from `tasks.json`, inside its `paths`.
- Import types from `@shared` only, and never invent tasks, fields, routes or packages.
- Use CSS Modules + tokens, and handle loading, error and empty states.
- Never commit secrets, and run `npm run check` before a PR.

To start an agent session with the right context, use `docs/ALLOCATOR.md`.

17. Common mistakes to avoid

Mistake	Instead
Writing <code>interface Country {...}</code> in a component	<code>import { type Country } from '@shared'</code>
Naming a type <code>Event</code>	<code>OrbitEvent</code>
<code>fetch('/api/countries')</code> inside a component	A hook in your feature folder using <code>apiGet(API_ROUTES.countries)</code>
Hard-coding <code> '/country/' + id</code>	<code>paths.country(id)</code> from <code>src/routes.ts</code>
<code>color: #00e5ff</code>	<code>color: var(--color-primary)</code>
<code>npm install some-lib</code> on your branch	Ask Hardik
Putting the AI key in code or a <code>VITE_</code> variable	Local <code>.env</code> only
Editing another person's folder "to fix a small thing"	Tell the owner
Country ID <code>in</code> , <code>India</code> or <code>ind</code>	<code>IND</code>
Dates like <code>19/09/2026</code>	<code>2026-09-19T00:00:00Z</code>
Deleting a failing test	Fix the code, or ask
Working inside OneDrive	<code>C:\dev\orbit</code>
A long-lived branch	PR every 2–4 hours

18. Definition of done

A task is done when:

- ☐ Every acceptance criterion in `tasks.json` is met
- ☐ The tests listed in the task exist and pass, and `npm run check` passes
- ☐ Only the task's `paths` changed
- ☐ Loading, error and empty states are handled, using tokens and primitives only
- ☐ The PR is merged into `main` by Hardik, who sets `status` to `done` in `tasks.json`
- ☐ The handoff message is posted in the team chat

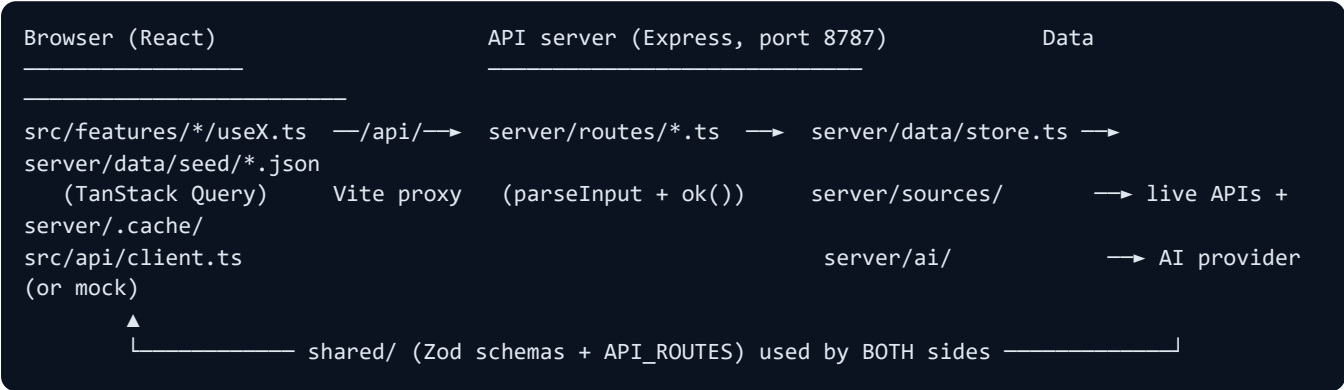
19. Personal LLM allocator

Each person pastes the prompt from `docs/ALLOCATOR.md` into their own LLM (Claude, ChatGPT, Cursor, Codex) with their name. It reads `tasks.json` and these docs, then tells them their role, tasks, branch, files, dependencies, acceptance criteria, tests and handoff. It never invents tasks.

ORBIT API and data contract

Owner: Hardik. The code is the truth: schemas live in `shared/schemas/`, paths in `shared/api.ts`, and routes in `server/routes/`. This document explains them in plain words. If you find a difference, the code wins, so tell Hardik and this file gets fixed.

How it fits together



- In development, the browser calls `http://localhost:5173/api/...` and Vite forwards it to `http://localhost:8787`.
- The browser never calls external APIs or AI directly. Keys stay on the server.

Response format (every route)

Success, HTTP 200:

```
{ "data": <the thing you asked for> }
```

Failure, HTTP 4xx/5xx:

```
{ "error": { "code": "NOT_FOUND", "message": "Country \"XXX\" not found" } }
```

code	HTTP	Meaning
VALIDATION	400	Bad input: wrong ID format, invalid body, invalid query
NOT_FOUND	404	The ID is valid but does not exist, or the route does not exist
UPSTREAM	502	A live source failed and there was no fallback (should be rare)
INTERNAL	500	A bug on the server (check the server terminal for <code>[api]</code> unexpected error)

In the frontend, `apiGet / apiPost` (`src/api/client.ts`) unwrap `data` for you and throw an `ApiClientError` (`code`, `status`, `message`) on failure. If the server is not running you get `code: "NETWORK"`.

Routes

Always build paths with `API_ROUTES` from `@shared`. The column "Builder" shows how.

Method + path	Builder	Returns (data)	Errors
GET /api/health	<code>API_ROUTES.health</code>	<code>HealthStatus { status: "ok", dataMode, aiProvider }</code>	none
GET /api/countries	<code>API_ROUTES.countries</code>	<code>Country[]</code>	none
GET /api/countries/:id	<code>API_ROUTES.country(id)</code>	<code>Country</code>	400 bad ID, 404
GET /api/countries/:id/events	<code>API_ROUTES.countryEvents(id)</code>	<code>OrbitEvent[]</code> newest first, every event whose <code>countryIds</code> includes <code>id</code>	400, 404
GET /api/countries/:id/timeline	<code>API_ROUTES.countryTimeline(id)</code>	<code>TimelineEvent[]</code> oldest first	400, 404
GET /api/events?kind=	<code>API_ROUTES.events(kind?)</code>	<code>OrbitEvent[]</code> newest first. <code>kind</code> optional: <code>geopolitical</code> or <code>health</code>	400 bad kind
GET /api/events/:id	<code>API_ROUTES.event(id)</code>	<code>OrbitEvent</code>	400 bad ID, 404
GET /api/news?countryId=	<code>API_ROUTES.news(countryId?)</code>	<code>NewsHeadline[]</code> newest first. Without <code>countryId</code> : all	400 bad ID
GET /api/sources	<code>API_ROUTES.sources</code>	<code>Source[]</code>	none
GET /api/insights/:subjectType/:subjectId	<code>API_ROUTES.insight(type, id)</code>	<code>AIInsight</code> . <code>subjectType</code> = <code>global</code> (<code>id</code> <code>world</code>), <code>country</code> (<code>country ID</code>) or <code>event</code> (<code>event ID</code>)	400 bad type, 404 unknown subject
POST /api/ask	<code>API_ROUTES.ask</code>	<code>AskAnswer</code>	400 invalid body

POST /api/ask body (`AskRequest`):

```
{
  "question": "Why is dengue rising?",
  "context": { "countryId": "IND", "eventId": "hs_ind_dengue_surge" }
}
```

`question` is 1–500 characters. `context` and both of its fields are optional. With `eventId`, ORBIT answers about that event. With only `countryId`, it answers about that country. With neither, it answers about the world.

Shared rules for all data

Rule	Example
Country ID = ISO 3166-1 alpha-3, uppercase	IND, USA, BRA (matches <code>properties.id</code> in <code>public/data/countries.geojson</code>)
Other IDs = prefix + lowercase slug (a-z, 0-9, _)	<code>evt_usa_tariff_review</code> , <code>hs_ind_dengue_surge</code> , <code>news_...</code> , <code>src_...</code> , <code>tl_...</code> , <code>ins_...</code>
Dates = ISO 8601 UTC with Z	<code>2026-09-19T10:00:00Z</code>
Coordinates = { lat, lng }	<code>{ "lat": 28.61, "lng": 77.21 }</code>
Severity = integer 1–5	1 Low, 2 Guarded, 3 Elevated, 4 High, 5 Critical
Field names = camelCase. Enum values = lowercase	<code>occurredAt</code> , <code>"geopolitical"</code>

Entities (data dictionary)

Country (`shared/schemas/country.ts`)

Field	Type	Notes
<code>id</code>	CountryId	IND
<code>name</code>	string	Display name
<code>region</code>	string	e.g. South Asia
<code>capital</code>	string	
<code>centroid</code>	{lat,lng}	Where the globe camera flies to
<code>population</code>	positive integer	
<code>summary</code>	string	One or two sentences
<code>riskLevel</code>	1–5	Overall current risk

OrbitEvent (`shared/schemas/event.ts`)

Either a **GeopoliticalEvent** (`kind`: "geopolitical", ID `evt_...`) or a **HealthSignal** (`kind`: "health", ID `hs_...`). Check `kind` before using kind-specific fields.

Field	Type	Notes
<code>kind</code>	geopolitical health	Decides which extra fields exist
<code>id</code>	evt_... or hs_...	Prefix must match <code>kind</code>

Field	Type	Notes
<code>title</code> , <code>summary</code>	string	
<code>countryIds</code>	CountryId[] (≥ 1)	The first one is the primary country (used for links)
<code>location</code>	{lat,lng}	Marker position
<code>occurredAt</code>	ISO date-time	
<code>severity</code>	1–5	
<code>sourceIds</code>	SourceId[]	Must exist in <code>sources.json</code>
<code>tags</code>	string[]	Free-form
geopolitical only: <code>category</code>	string	Free text until the category list is agreed
geopolitical only: <code>actors</code>	string[]	Governments, organisations, people
health only: <code>indicator</code>	string	e.g. <code>Dengue cases</code>
health only: <code>metric</code> (optional)	{ value: number, unit: string } }	e.g. { 18400, "cases this month" }

NewsHeadline (`shared/schemas/news.ts`)

`id` (`news_...`), `title`, `url`, `sourceId`, `publishedAt`, `countryIds` (may be empty), `eventId` (optional, links to an OrbitEvent).

Source (`shared/schemas/source.ts`)

`id` (`src_...`), `name`, `url`, `type` (`news` | `government` | `international_org` | `research` | `dataset`), `reliability` (`high` | `medium` | `low`).

TimelineEvent (`shared/schemas/timeline.ts`)

`id` (`tl_...`), `countryId`, `date`, `datePrecision` (`day` | `month` | `year`), `title`, `description`, `eventId` (optional). A year-only fact is stored as 1 January of that year with `datePrecision`: "year" .

AllInsight (`shared/schemas/insight.ts`)

`id` (`ins_...`), `subjectType` (`global` | `country` | `event`), `subjectId` (`world`, a country ID or an event ID), `summary`, `keyPoints` (≥ 1), `sourceIds`, `confidence` (`low` | `medium` | `high`), `provider` (`mock`, `google`, ...), `generatedAt` . It is explainable because it always says which sources it used, how sure it is, and which AI produced it.

AskRequest / AskAnswer (`shared/schemas/ask.ts`)

Request: `question`, optional `context` { `countryId?`, `eventId?` } . Answer: `answer`, `sourceIds`, `provider`, `generatedAt` .

Relationships

```
Country (IND) ← countryIds — OrbitEvent (evt_/hs_) — sourceIds → Source (src_)
    ↑           ↑           ↑
    | countryId | eventId — NewsHeadline (news_) — sourceId →
Source
    |
    |— TimelineEvent (tl_) — eventId (optional)
AIInsight (ins_) — subjectId → world | Country | OrbitEvent, sourceIds → Source
```

The server checks every one of these links when it starts, and `npm test` checks them too (`server/data/seed.test.ts`). A broken link stops the server with a message such as `news.json news_x: unknown source "src missing"`.

Mock data format (server/data/seed/)

One JSON file per entity, each a plain **array** of objects matching the schema above:

File	Contains	Owner
countries.json	Country[]	Shrey
events.json	OrbitEvent[] (both kinds mixed, each with kind)	Shrey
news.json	NewsHeadline[]	Shrey
sources.json	Source[]	Shrey
timeline.json	TimelineEvent[]	Shrey

The current files are **small illustrative samples** (3 countries, 6 events). Everything marked "Mock" and every `src_mock_*` source is placeholder content, to be replaced with real, sourced data (tasks ORB-SHR-01 and ORB-SHR-02). To check your edits, run `npm test`. The seed test prints the exact file, item and field that is wrong.

Data modes and future live data

Setting	mock (default)	live
DATA_MODE	Seed data only, fully offline	Seed data + live sources (server/sources/)
AI_PROVIDER	Canned answers from seed data	google (Gemini). Falls back to mock on any failure

Rules for every live source (server-side only):

1. Fetch → validate with the shared schema (drop invalid items) → save with `writeCache(key, data)`.
2. Merge with seed data: seed is always the base, live adds to it.
3. On any failure: log `[sources] ...` and return `readCache(key)`, then seed. Never throw to the route.
4. Live items must follow the same ID rules (e.g. `news_<source>_<slug>`) and reference existing sources.
5. A new external API key = a new empty line in `.env.example` + a field in `server/env.ts`.

Changing this contract

Adding a field, route or entity is a contract change. Follow [docs/INTEGRATION.md#contract-changes](#) . Hardik updates [shared/](#) , this document and the tests in one PR, which merges before the feature that needs it.

ORBIT integration contract

Owner: Hardik. This document describes how five branches become one working ORBIT app without breaking each other.

The flow

```
main → your branch (from tasks.json) → commits → pull main + npm run check → PR → Hardik merges (squash) → main
```

1. `git checkout main && git pull` then `git checkout -b <branch from tasks.json>` (e.g. `ayman/country-view`).
2. Work only inside your task's `paths` (see `tasks.json` and `AGENTS.md` section 3).
3. Commit small and often.
4. Before the PR: `git pull origin main` (merge main into your branch), resolve conflicts, `npm run check`.
5. Open a PR into `main`. Hardik reviews, merges (squash), and posts in the chat.
6. Everyone: `git checkout main && git pull && npm install`, then start the next task on a new branch.

Branch naming

`<lowercase name>/<short-topic>`, exactly as in the task's `branch` field. Examples: `arham/globe`, `affan/google-provider`. Use one branch per task and delete it after merge. Never commit directly to `main`.

Commits

- Format: `<area>: <what changed>` in the present tense, e.g. `country: show risk level`, `seed: add 10 countries`.
- One logical change per commit. Never commit generated or secret files (see "Never commit").

Pull requests

- Title: `[ORB-XXX-NN] short summary`, e.g. `[ORB-AYM-01] Country view`.
- The description says: what changed, how to test it (the click path), the tests you added, and any contract/API changes (these should already be merged separately), plus screenshots for UI.
- Keep PRs small. A PR every 2–4 hours is ideal. A PR that touches another owner's folder is sent back.
- Checklist before requesting a merge:
 - ☐ `npm run check` passes
 - ☐ Only my task's `paths` changed (plus tests)
 - ☐ Loading / error / empty states handled
 - ☐ No secrets, no `.env`, no new packages (unless Hardik added them)

Shared-file rules

Shared hot files (AGENTS.md section 3): `shared/**`, `package.json`, `package-lock.json`, `src/App.tsx`, `src/routes.ts`, `src/styles/tokens.css`, `src/ui/index.ts`, `server/app.ts`, `.env.example`, `tasks.json`, `AGENTS.md`.

- Only their owner edits them, in a **separate small PR** that is merged **before** the feature PR that needs it.
- Everyone else **imports** from them freely.

Contract changes

A contract change is any change to `shared/` (a field, schema, enum, ID rule, route or response shape).

1. **Ask** in the team chat: `CONTRACT CHANGE: add optional imageUrl to Country, needed for ORB-AYM-01`.
2. **Hardik** makes the change in one PR: `shared/` + `docs/API.md` + the server route or seed validation + tests. New fields start as `.optional()`, so existing data and code keep working.
3. Hardik merges it and posts `CONTRACT UPDATED: ... pull main`.
4. Everyone pulls main. Then the feature PR uses the new field.

Never change the meaning of an existing field and never delete a field while someone uses it. Add a new field instead.

API changes

Same process as contract changes. New routes are added to `API_ROUTES` (`shared/api.ts`) and the routes table in `docs/API.md` in the same PR as the server code. Frontend code uses a new route only after that PR is on `main`.

How conflicts are avoided

- Folder ownership (AGENTS.md section 3): two people should never edit the same file.
- All routes are declared up front in `src/App.tsx`, so features never touch the router.
- Each feature has its own query hooks and keys. There is no shared query key file.
- Only Hardik installs packages, so there are no lockfile conflicts.
- CSS Modules per component, so there are no global CSS conflicts. Only Arham edits `tokens.css`.
- Seed data: only Shrey edits `server/data/seed/`.

If you do hit a conflict in a file you don't own, **don't resolve it yourself**. Ask the owner.

Testing before merging

The author runs `npm run check` (lint, tests, typecheck, build) and clicks through their change in `npm run dev`. Hardik runs `npm run check` on `main` after each merge batch and clicks Global → Country → Event → Ask (task ORB-HAR-02). If `main` breaks, fixing it comes before any new work. Revert the PR if it can't be fixed in 15 minutes.

Sync points

About every 4 hours (task ORB-HAR-02), everyone pushes what works, Hardik merges in order (`shared/` changes first, then server, then UI), and everyone pulls. Keep your branches short-lived between sync points.

Never commit

- `.env` or any real key or token (only `.env.example` with empty values)
- `node_modules/`, `dist/`, `server/.cache/`
- Another lockfile (`yarn.lock`, `pnpm-lock.yaml`, `bun.lockb`)
- Files outside your task's paths, or edits to `mcp/` and `workshop-data/`
- Commented-out code blocks, `console.log` debugging left behind, disabled tests or lint rules

ORBIT: rules for coding agents and humans

This file is the single source of rules for Claude Code, Codex, Cursor and people. `CLAUDE.md` and `.cursor/rules/orbit.mdc` only point here. If another document disagrees with this file, this file wins. Tell Hardik so the other document gets fixed.

0. Before you change anything

- 1. Know **who you are working for** and **which task** (`tasks.json` , e.g. `ORB-AYM-01`). No task, no code.
- 2. Only edit files inside that task's `paths` and the owner's folders (section 3).
- 3. Run `npm run check` before every commit you intend to push.

1. What ORBIT is

An AI-powered global intelligence platform on a 3D globe: **Global** → **Country** → **Event** → **Explanation** → **Ask ORBIT**. It's a 24-hour hackathon prototype, demoed live on one laptop, and it must work offline.

2. Stack (frozen, do not change or add alternatives)

Area	Choice
Frontend	Vite 8 + React 19 + TypeScript (strict)
3D globe	react-globe.gl (three.js)
Server	Express 5, run with tsx, in <code>server/</code>
Contracts	Zod 4 schemas in <code>shared/</code> , imported as <code>@shared</code>
Data fetching	TanStack Query (hooks live in each feature folder)
UI state	Zustand (<code>src/state/</code>). The selection lives in the URL (<code>src/routes.ts</code>)
Styling	CSS Modules + tokens in <code>src/styles/tokens.css</code>
AI	Vercel AI SDK behind <code>server/ai/</code> , with the mock provider as default and fallback
Data	JSON seed files in <code>server/data/seed/</code> + disk cache <code>server/.cache/</code> . No database
Tests	Vitest + Testing Library
Package manager	npm only (one lockfile: <code>package-lock.json</code>). Node ≥ 22.12

Do not introduce Next.js, Tailwind, Redux, a database, Python services, another package manager, or another app.

3. Folder ownership

Only the owner edits these folders. Everyone may **import** from anywhere, but must not **edit** outside their folders.

Owner	Owns
Hardik (integrator)	shared/, server/app.ts, server/index.ts, server/env.ts, server/http.ts, server/testServer.ts, server/routes/ (except ai.ts), server/data/store.ts, server/sources/, src/App.tsx, src/routes.ts, src/main.tsx, src/queryClient.ts, src/api/, src/test/, scripts/, root config files, package.json, package-lock.json, .env.example, tasks.json, AGENTS.md, CLAUDE.md, .cursor/, README.md, docs/ (except the files below)
Affan	server/ai/, server/routes/ai.ts, src/features/ask/, src/features/insight/
Arham	src/styles/, src/ui/, src/globe/, src/layout/, src/state/, public/, docs/DESIGN_SYSTEM.md
Ayman	src/features/country/, src/features/event/, src/features/timeline/
Shrey	server/data/seed/, src/features/news/, src/features/health/, docs/DEMO.md

Never touch: mcp/, workshop-data/, .vscode/mcp.json (hackathon organiser tooling).

Shared hot files (only change through a small, separate PR that Hardik merges first): shared/**, package.json, package-lock.json, src/App.tsx, src/routes.ts, src/styles/tokens.css, src/ui/index.ts, server/app.ts, .env.example, tasks.json, AGENTS.md.

4. Shared contracts (shared/)

- Every data type is defined **once**, as a Zod schema in shared/schemas/: XSchema + type X = z.infer<typeof XSchema>.
- Import types and schemas only from @shared: import { type Country, API_ROUTES } from '@shared'.
- **Never** write your own interface Country, type Event, etc. **Never** name a type Event (it clashes with the DOM). Use OrbitEvent.
- IDs: countries = ISO alpha-3 uppercase (IND). Everything else = prefix + lowercase slug: evt_ geopolitical event, hs_ health signal, news_, src_, tl_, ins_.
- Dates: ISO 8601 UTC strings (2026-09-19T10:00:00Z). Coordinates: { lat, lng }. Severity: integer 1–5.
- Fields: camelCase. Enum values: lowercase strings.
- Changing a contract: follow docs/INTEGRATION.md#contract-changes. New fields start as .optional().

5. API rules

- All routes are listed in docs/API.md and built with API_ROUTES from @shared. Never hand-type /api/... strings.
- Success responses: { "data": ... }. Error responses: { "error": { "code", "message" } } plus an HTTP status.
- Server: validate every input with parseInput(schema, value). Throw AppError / notFound() (in server/http.ts). Never res.status(...).json(...) by hand.

- Frontend: the only `fetch` calls live in `src/api/client.ts`. Features call `apiGet` / `apiPost` inside TanStack Query hooks in their own folder.
- Query keys start with the feature name (`['country', id]`) so features never collide.
- Only the server talks to external APIs or AI providers. The browser never does.

6. Environment and secrets

- Real values go in a local `.env` (gitignored). `.env.example` lists every variable with a safe empty or default value.
- **Never** commit keys, tokens or `.env`. **Never** put a secret in a `VITE_` variable (those ship to the browser).
- Adding a variable: add it to `.env.example` **and** `server/env.ts` in the same PR (Hardik merges).

7. Dependencies

- Do not run `npm install <package>`. Ask Hardik in the chat, with the package name and why. Hardik installs, commits the lockfile, and everyone pulls.
- Everyone else only runs plain `npm install` (after pulling) to sync.
- Never delete or regenerate `package-lock.json`. Never use yarn, pnpm or bun.

8. Styling rules

- CSS Modules only: `Component.module.css` next to `Component.tsx`. No global class names except in `src/styles/global.css`.
- Use tokens: `var(--color-primary)`, `var(--space-4)` ... **No raw hex colours, px font sizes or shadows outside** `tokens.css`.
- Canvas/WebGL code reads tokens with `cssVar()` from `src/styles/cssVar.ts`.
- Reuse primitives from `src/ui` (Panel, Loader, ErrorState, ErrorBoundary, QueryState, ItemList, SeverityBadge). Need a new one? Ask Arham.
- Details: `docs/DESIGN_SYSTEM.md`.

9. Error handling

- Every query renders loading **and** error states. Use `<QueryState query={...} label="...">`.
- Wrap each feature section in `<ErrorBoundary title="...">` so one crash never blanks the app.
- Server: never crash on bad data from a live source or AI. Log `[sources] ... / [ai] ...` and fall back (cache → seed / mock).
- Logging uses a prefix: `[api]`, `[sources]`, `[ai]`, `[ui]`.

10. Testing

- `npm run check` = lint + tests + typecheck/build. It must pass before every PR.
- Each task lists the tests it needs in `tasks.json`. Put tests next to the code (`x.test.ts(x)`).
- Server/shared tests start with `// @vitest-environment node`. Frontend tests use `renderWithProviders` from `src/test/utils.tsx`.
- The globe is mocked in tests (no WebGL in jsdom). Do not remove that mock.

11. Git workflow

- `main` must always run. Only Hardik merges (Affan is backup).
- Branch per task: the `branch` field in `tasks.json` (`<name>/<topic>` , e.g. `arham/globe`).
- Commit messages: `<area>: <what changed>` (e.g. `globe: add severity rings`). Small commits.
- Before opening a PR: `git pull origin main` , fix conflicts in YOUR files only, run `npm run check` .
- PR title: `[ORB-XXX-NN] short summary` . Squash merge.
- Full rules: `docs/INTEGRATION.md` .

12. What agents must NOT do

- Invent tasks, features, routes, fields or packages that are not in `tasks.json` / `docs/API.md` .
- Edit files outside the current task's owner folders "while they're at it".
- Change the stack, add a framework, or restructure folders.
- Disable lint rules, delete tests or loosen TypeScript to make `npm run check` pass.
- Commit `.env` , keys, `dist/` , `server/.cache/` or `node_modules/` .

ORBIT design system

Owner: Arham. The tokens live in `src/styles/tokens.css` (the only place raw values may appear). The primitives live in `src/ui/`.

Direction: deep black, electric cyan, amber/orange accents. Futuristic, cinematic, clean, like an intelligence centre. The globe is the hero, and panels float over it as dark glass.

Tokens

Colour

Token	Use
<code>--color-bg</code> , <code>--color-bg-elevated</code>	Page background, raised areas
<code>--color-surface</code> , <code>--color-surface-hover</code>	Glass panels (semi-transparent over the globe)
<code>--color-border</code> , <code>--color-border-strong</code>	Hairlines, focused/active outlines
<code>--color-text</code> , <code>--color-text-muted</code> , <code>--color-text-faint</code>	Primary, secondary, tertiary text
<code>--color-primary</code> , <code>--color-primary-dim</code>	Electric cyan: links, focus, geopolitical
<code>--color-accent</code>	Amber: Ask ORBIT, health, highlights
<code>--color-alert</code>	Orange: critical, errors
<code>--color-success</code> , <code>--color-danger</code>	Status only
<code>--severity-1</code> ... <code>--severity-5</code>	The shared 1–5 severity scale (cyan → amber → orange)
<code>--kind-geopolitical</code> , <code>--kind-health</code>	Event kind colours
<code>--globe-*</code>	Ocean, land, hover, selected, borders, atmosphere

Typography

Token	Value
<code>--font-display</code>	Space Grotesk: headings, brand
<code>--font-body</code>	Inter: body text
<code>--font-mono</code>	JetBrains Mono: labels, numbers, badges
<code>--text-xs</code> ... <code>--text-3xl</code>	0.75rem → 2.75rem scale
<code>--weight-regular/medium/bold</code>	400 / 500 / 700
<code>--tracking-label</code>	Letter spacing for uppercase eyebrow labels
<code>--leading-tight</code> , <code>--leading-body</code>	Line heights

Fonts load from Google Fonts. Offline, they fall back to Segoe UI / Consolas, so the demo still reads well.

Spacing, borders, radii

- Spacing on a 4px grid: `--space-1` (4px) ... `--space-8` (64px).
- `--border-width` (1px). Radii: `--radius-sm` 4px, `--radius-md` 8px, `--radius-lg` 14px (panels), `--radius-pill`.

Shadows and glows

`--shadow-panel` (panel depth), `--glow-primary` (cyan), `--glow-accent` (amber), `--glow-alert` (orange), `--blur-panel` (glass blur). Use glows sparingly: for focus, hover and severity 5 only.

Motion and layers

- `--duration-fast/base/slow` (150/250/600ms) with `--ease-out`. They become 0 when the user prefers reduced motion.
- `--z-globe` < `--z-overlay` < `--z-panel` < `--z-modal`.
- Layout sizes: `--panel-width` (420px), `--topbar-height` (56px).

Primitives (`src/ui` , import from `../..ui`)

Component	Use
<code>Panel</code>	Every overlay section. Props: <code>eyebrow</code> , <code>title</code> , <code>actions</code> , <code>children</code>
<code>QueryState</code>	Renders loading / error / data for a TanStack query: <code><QueryState query={q} label="country"> {(data) => ...}</QueryState></code>
<code>Loader</code>	Loading state (<code>role="status"</code>)
<code>ErrorState</code>	Error state with optional retry (<code>role="alert"</code>)
<code>ErrorBoundary</code>	Wrap each feature section so a crash stays contained
<code>ItemList</code>	Vertical list with dividers and a required empty state
<code>SeverityBadge</code>	Shows severity 1–5 with its colour and label

Button and Tag are planned (task ORB-ARH-01). Ask Arham before creating a new primitive.

Common UI states (every data view needs all four)

State	How
Loading	<code>QueryState</code> / <code>Loader</code>
Error	<code>QueryState</code> / <code>ErrorState</code> (with retry)
Empty	<code>ItemList</code> <code>empty</code> text, or a sentence in muted text
Success	The content

Conventions

- One `Component.module.css` next to each component. Class names are camelCase (`styles.askButton`).
- Only tokens: `color: var(--color-text-muted)` , never `#8aa0b8` . The same goes for spacing, radii and shadows.
- Canvas/WebGL (globe): read tokens with `cssVar('--globe-land')` and `severityColor(severity)` from `src/styles/cssVar.ts` .
- Text: sentence case. Eyebrow labels are uppercase via CSS, not typed in capitals.
- Accessibility: visible focus (global `:focus-visible` ring), `aria-label` on icon-only buttons, and colour never the only signal (severity badges also show the label and number).
- Numbers: `toLocaleString()` for large values. Dates: show them at their `datePrecision` .

ORBIT personal LLM allocator

Paste the prompt below into your own LLM (Claude Code, Codex, Cursor, ChatGPT...). Replace `<YOUR NAME>` with one of: **Arham, Ayman, Affan, Shrey, Hardik**. If your LLM cannot read files from the repository, paste the contents of `tasks.json`, `AGENTS.md`, `docs/PLAYBOOK.md`, `docs/API.md` and `docs/INTEGRATION.md` after the prompt.

You are my personal task allocator for the ORBIT hackathon project. My name is: <YOUR NAME>

SOURCES (the only truth, read all of them before answering):

1. tasks.json - the authoritative task list
2. AGENTS.md - rules, folder ownership, shared hot files
3. docs/PLAYBOOK.md - overview, ownership table, dependency graph, definition of done
4. docs/API.md - API routes and data contracts
5. docs/INTEGRATION.md - branches, commits, PRs, contract changes
6. docs/DESIGN_SYSTEM.md - only if my tasks touch UI

HARD RULES:

- Use ONLY tasks from tasks.json whose "owner" is exactly my name. Never invent, merge, split or rename tasks.
- Never assign me files outside my task "paths" and my folders in AGENTS.md section 3.
- If something I ask for is not in tasks.json, say "Not in tasks.json - ask Hardik" instead of making it up.
- If my name is not one of Arham, Ayman, Affan, Shrey, Hardik, stop and ask me for the correct name.
- Quote IDs, paths, branches and acceptance criteria exactly as written in the sources.

OUTPUT (in this order, concise, using headings):

1. My role: the "team" entry for me in tasks.json, plus my row from the ownership table in docs/PLAYBOOK.md.
2. My tasks: a table of all my tasks sorted by priority (P0, P1, P2), then targetWindow, with:
ID | title | priority | targetWindow | status | branch | dependsOn (and whether each dependency is done).
3. Start now: the single task I should do next (highest priority, earliest window, all dependsOn done, status not done). If every task is blocked, say which dependency I am waiting for and who owns it.
4. For that task:
 - Description (verbatim)
 - Branch name and the exact git commands to create it from an up-to-date main
 - Files and folders I may edit (task "paths" plus my owned folders)
 - Files I must avoid: other people's folders, the shared hot files from AGENTS.md section 3 (unless I own them),
mcp/, workshop-data/, .env
 - Expected inputs: which contracts from @shared, which API routes (docs/API.md), and which other people's components or data I rely on
 - Expected outputs: what must exist when I am done
 - Acceptance criteria (verbatim) as a checklist
 - Tests required (verbatim) plus "npm run check passes"
 - Commit and PR instructions: commit message format, PR title "[<task id>] <summary>", the PR checklist from docs/INTEGRATION.md
 - Integration handoff: the task's "handoff" text, and who to tell
5. Rules I must remember: the 6 most relevant rules from AGENTS.md for this task.
6. After this task: what to do next (my next task by the same ordering, or "tell Hardik you are free").

Then ask me whether I want step-by-step help implementing the "Start now" task. While helping, obey AGENTS.md and stay inside the task's paths.

How the allocator stays correct

- It only reads the files above, so updating `tasks.json` (through a PR merged by Hardik) updates every person's allocation.

- `npm test` validates `tasks.json` (unique IDs, valid owners, existing dependencies, no cycles, everyone has a task).
- When Hardik merges your PR, they set the task `status` to `done` in `tasks.json`, so the allocator moves you on. Pull main to see it.